



Spreadsheet error and fixes

Hi and welcome back. Recently we've been learning about formulas. Sometimes data analysts encounter a problem with our formulas and we get an error. We've all been there and it can be frustrating. But there are solutions, that's what we're going to explore in this video. One error you may encounter is the DIV error. The DIV error happens when a formula is trying to divide a value in a cell by zero or by an empty cell. In this spreadsheet, the percentage Complete values in column C are calculated by dividing the values in the Tasks Completed column by the values in the Required Tasks column. Notice that column C is already formatted as a percentage. The DIV error is in cell C4 because we're dividing by zero the value in cell A4. To avoid this problem, we can have this spreadsheet automatically enter not applicable whenever a cell in column A contains a zero that would cause the error. To do this, we'll use the IFERROR function. If it encounters a DIV error caused by a cell that contains the zero, the phrase "Not applicable" will be inserted. We can also copy the formula to the rest of the cells in column C so it checks for any other cells that contain a zero. Now let's move on to ERROR. In Google Sheets, ERROR tells us the formula can't be interpreted as it is input. This is also known as a parsing error. Say we want to tally the number of total tasks in column B and C, we use the SUM function, but the formula equal sum B2 to B6, C2 to C6 causes an error. Examining it more closely, we see that a comma is missing between the cell ranges B2 to B6 and C2 to C6. We can fix this by inserting a comma between the cell ranges to indicate the end of each data item. This is called a delimiter, which you will learn more about soon. Now, the formula can correctly calculate the total number of tasks as 25. Another type of error is N/A. The N/A error tells you that the data in your formula can't be found by the spreadsheet. Generally, this means the data doesn't exist. This error most often occurs when using functions such as VLOOKUP, which searches for a certain value in a column to return a corresponding

piece of information. Here, we see a master list of nuts and their prices. Using VLOOKUP, the spreadsheet finds prices in the list, then calculates the prices for each store using the assigned markup. But we have a N/A error in cells B49 and C49. The VLOOKUP formula is correct, so what's going on? Well, if we look carefully at the name of the nut, "almond" has no match in the lookup table, the lookup table uses the plural "almonds" instead. So we change almond to almonds, and with that typo fixed, the right prices are filled in. Speaking of typos, sometimes a typo can cause a **NAME error**. **A NAME error can happen when a formula's name isn't recognized or understood**. Suppose we see a NAME error in the nut prices spreadsheet. If we look carefully, the VLOOKUP function in cell B21 is spelled incorrectly, it has one extra O; this causes a NAME error for both the price and the resulting markup calculation for the store. To fix this error, we can delete the extra O in VLOOKUP. Perfect. Sometimes an error is caused by inconsistent or wrong data. For instance, the NUM error tells us that a formula's calculation can't be performed as specified by the data. The data doesn't make sense for that calculation. Here's what I mean. Suppose we're working on a large construction project using a spreadsheet to track how many months it takes to reach key milestones. We can use the DATEDIF function to calculate the number of months between start and end dates. The function requires the start date to be in the first cell referenced and the end date to be in the second cell referenced. In our case, cells B2 and C2 respectively. The M represents months, as we want this spreadsheet to calculate the number of months between our start and end dates. But we get a NUM error in cell D6. We notice that the end date comes before the start date, so the DATEDIF function can't calculate the number of months between. It's likely the start and end dates were interchanged by accident. We can request verification of the data to make sure. In the meantime, let's reverse the order of the cells in the formula to temporarily get around the error. Now, the result is nine months. What if the client's name was accidentally inserted into the start date in the spreadsheet? You guessed it, we get an error. **The VALUE error can indicate a problem with a formula or referenced cells**. It's often not clear right away what the problem is, so this error might take a little more effort to fix. In this case, John Welty was input as the start date, making the calculation

impossible for the DATEDIF function in the cell D6. We just replace the text, John Welty, with the correct start date of September 1st, 2016. Last is the REF error, which often comes up when cells being referenced in a formula have been deleted, thus making the formula unable to perform the calculation. Here's a spreadsheet

used to calculate the number of seats available for a company lunch. Let's say the company decided not to run the second floor, so we delete row 4. This results in a REF error when calculating the total seats available in cell B5. To fix this, we can change the formula to add the values in cells B2 and B3. Also, in this case, we could have prevented the REF error by using the SUM function and a range of cells instead of adding the cell value by direct reference. Now, if we delete row 10, the SUM function calculates the total seats available. There you go. We've now fixed some of the most common spreadsheet errors. When you see them again, you'll know what they mean. Troubleshooting is a big part of data analysis, so being able to find solutions is a key skill for data analysts.